# Sign & Say — Smart Home Mobility Project

System architecture & data flow · all AI runs on the Arduino VENTUNO Q (Qualcomm QCS8275) · the laptop only displays the UI

## LOGITECH CAMERA

USB, plugged into the board

**Image sensor**
1280×720 MJPEG
30 fps · /dev/video0

USB video

## ARDUINO VENTUNO Q

Qualcomm QCS8275 SoC · Kryo CPU (Ubuntu 24.04) · Hexagon NPU HTP V75 · on-board MCU + LED matrix

**BUILD TIME — PyTorch → ExecuTorch export (run once, on the board)**

- **PyTorch 2.13 models** — MediaPipePyTorch (AI Hub pin)
- **torch.export** — capture the graph
- **QnnQuantizer 16a8w** — calibrated on real frames
- **QnnPartitioner** — target QCS8275 · HTP V75
- **.pte files (export/pte/)** — 100% NPU-delegated · 0.5–3.4 MB

deploy .pte models to the NPU

MJPEG video (passthrough, no re-encode)

21 keypoints / hand → /hands.sse

**Camera capture** — OpenCV V4L2, latest-frame thread

**Hand tracker** — hand_tracker.py, letterbox · crops · NMS

**Face filter** — drop hands with >60% keypoints in a face

**One Euro filter** — smooth 21 keypoints, up to 2 hands

**Gesture rules 👍 👎** — gestures.py, 3D thumb + fist geometry

**GestureState** — 5 of 10 frames, 3 s hold · instant switch

### HAND GESTURE PIPELINE

Home server · camera_stream.py :8090
Kryo CPU · Python 3.12 · NumPy + OpenCV
started at boot by cron (run_home_server.sh)

frames → NPU
← boxes / keypoints

face boxes

**Voice + gesture fusion** — 'light' + 👍 /👎 within 3 s → lights on / off

**Home state** — Lighting · Temperature °F · TV, home_state.json

voice 'light' → /api/intent → wait for 👍 / 👎

### QUALCOMM HEXAGON NPU (HTP V75)

**ExecuTorch qnn_executor_runner — 1 persistent process per model**
QNN context stays loaded · frames in/out via /dev/shm

- **Palm detector** — MediaPipe BlazePalm, 16a8w · 1.42 ms (CPU: 116 ms)
- **Hand landmark** — 21 × 3D keypoints, 16a8w · 1.39 ms (CPU: 75 ms)
- **Face detector** — MediaPipe BlazeFace, 16a8w · 1.3 ms, every 2nd frame

**Arduino App Lab inference containers**
ei-qnn-models-runner · audio-analytics (QNN)

- **Wake-word model** — Edge Impulse KWS, 'Hey Arduino' keyword spotting
- **Whisper Small** — speech → text, English, fully offline

NPU software stack

- **QNN delegate** — ExecuTorch backend
- **QAIRT / QNN 2.49**
- **FastRPC (CPU ↔ DSP)**
- **Hexagon HTP V75** — tensor cores

### VOICE AI PIPELINE

Arduino App Lab app 'hey-arduino-home' · runs in Docker containers

**Microphone + pre-roll** — PrerollMicrophone, 8 s rolling buffer

audio stream

**KeywordSpotting brick** — 'Hey Arduino' ≥ 0.90, 'Hey Ardy' ≥ 0.30

**ASR brick (Whisper)** — asr_session.py, 6 s deadline flush

**commands.py** — wake confirm + parse, lights · temp · TV

**On-board MCU** — sketch.ino via Bridge, LED matrix: listening, done, not understood

**Arduino App Lab platform**
arduino-app-cli 0.13 · bricks: arduino:keyword_spotting, arduino:asr
containers: python-apps-base 0.12 · ei-qnn-models-runner 0.12.1
audio-analytics 1.0.2

## Built-in microphone

ALSA capture
always listening

USB audio

## HTTP API

port 8090

/stream.mjpg
/hands.sse
/gesture
/api/events
/api/state
/health

from voice app:
/api/state
/api/intent
/api/voice

also bound on docker0 for App Lab containers

USB-C

## MACBOOK LAPTOP

UI only — no AI runs here

**adb forward tcp:18090 → board :8090** — over the USB-C cable

**ventuno_ui.py → http://127.0.0.1:9999** — launchd agent com.ventuno.home-ui, USB watcher · deploys scripts · HTTP proxy

**Browser — index.html** — 'Sign & Say' house UI

- **Camera view** — live MJPEG + hand skeleton overlay, 👍 / 👎 gesture badge
- **Lighting — on/off, brightness**
- **Temperature — actual | target °F**
- **TV — Comedy · Sci Fi · Action**
- **Status: board · camera fps · gesture · voice**
- **hls.js player** — TV stream inside the camera window

**Pluto TV (internet)** — HLS live channel streams

Logitech camera unit · Arduino VENTUNO Q board · Hand gesture pipeline (CPU) · Qualcomm Hexagon NPU · Voice AI (Arduino App Lab) · Build time (PyTorch → ExecuTorch) · Laptop UI

live data · request / response · deploy / external

**Hand tracking ~30 fps · NPU: palm 1.42 ms, landmarks 1.39 ms, face 1.3 ms (CPU fp32: 75–116 ms) · Voice: wake word + Whisper on the NPU, no cloud**